

## MA 550 Statistical Data Analysis

**Course Code: MA 550**

**Course Name: Statistical Data Analysis**

L-T-P-C: 2-1-0-3

Students intended for: MS/ Ph.D., Undergraduate (3rd and 4th year)

Prerequisites: MA 202 for undergraduate

Elective or Core: Elective

Approval: 2nd Senate

### Course Contents:

- **Exploratory analysis of time series:**

Introduction, examples, simple descriptive techniques, trend, seasonality, stochastic and deterministic approaches; numerical and experimental data sets; challenges in data analysis and data graphical representation, interpretation; statistical tests, significance and power of a test, choice of the critical region, constructing test statistics: the Fisher discriminant, mean and variance test, testing goodness-of-fit, chi2-test, p-values; stationary time series process (ARMA Processes). [12 Lectures]

- **Analysis of stochastic series:**

Model identification and non-stationary time series models; forecasting with classical regression models; Forecasting with autocorrelations; Forecasting with lagged dependent variable; Forecast error statistics and evaluation; singularity detection, spectral density function, the periodogram, spectral analysis, correlogram, wavelet cross-correlation, multi-resolution analysis, examples and applications. [12 Lectures]

- **Clustering data techniques:**

Principal component analysis; different techniques of data clustering. [6 Lectures]

### Text books:

1. Peck and Devore, **Statistics: The Exploration and Analysis of Data**, 7th edition, Thomson-Brooks/Cole, 2012.
2. Montgomery, D., Jennings, C.L. and Kulahci, M., **Introduction to Time Series Analysis and Forecasting**, Wiley-Interscience, 2008.
3. Chatfield, C., **The Analysis of Time Series**, 6th Edition, Chapman & Hall/CRC, 2004.

### References:

1. Petre Stoica and Randolph L. Moses, **Introduction to Spectral Analysis**, Prentice Hall, 1997.
2. Robert H. Shumway and David S. Stoffer, **Time Series Analysis and Its Applications with R Examples**, 3rd edition, Springer Texts in Statistics, 2006.
3. Raghuveer M. Rao and Ajit S. Bopardikar, **Wavelet Transform**, Pearson Education, 1998.